Trends and Cycles I

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Contents

1 Recap

- Most macroeconomic data are time series — thus we need to know about time series fundamentals
  - The stationarity concept
  - Basic building blocks — AR and MA
- A simple theory model helpful for making you think logically consistent about macroeconomic fluctuations
  - But variables are often framed as gaps (cycles), or at least stationary variables
- Most macroeconomic data are not stationary
  - Different "rules" apply for data that is nonstationary
- Trends often make data nonstationary
  - Removing the trend, or modelling the trend directly, allows us to use the stationary "rules"
- To do this correcly, it is important to separate between different trend processes:
  - Deterministic trends
  - Stochastic trends

1.1 Nonstationarity defined

A time series process where either the first- or second-order moments (that is, mean and variance, respectively) depend on the time index is nonstationary

2 Deterministic Trends

Definition:

If a process includes t, it contains a deterministic trend

- consider the moving average process

$$y_t = \delta + \alpha t + \theta(L)\epsilon_t$$

  where $\theta(L) = 1 + \theta_1 L + \theta_2 L^2 \dots$ and the deterministic trend is simply the time index t, with a slope parameter α
- Including the trend, makes the process nonstationary:

$$\mathbb{E}[y_t] = \delta + \alpha t$$

  i.e., the mean depends on time
- Note that deviations of y_t from its mean are stationary:

$$\begin{aligned} y_t - \mathbb{E}[y_t] &= \delta + \alpha t + \theta(L)\epsilon_t - (\delta + \alpha t) \\ &= \theta(L)\epsilon_t \end{aligned}$$

- i.e., by removing the deterministic trend, the process becomes stationary

3 Stochastic Trends

- The counterpart of a deterministic trend is a stochastic trend
- The simplest model to think of with a stochastic trend is a Random Walk (RW):

$$y_t = y_{t-1} + \epsilon_t \sim \text{i.i.d. } N(0, \sigma^2)$$

3.1 RW Properties

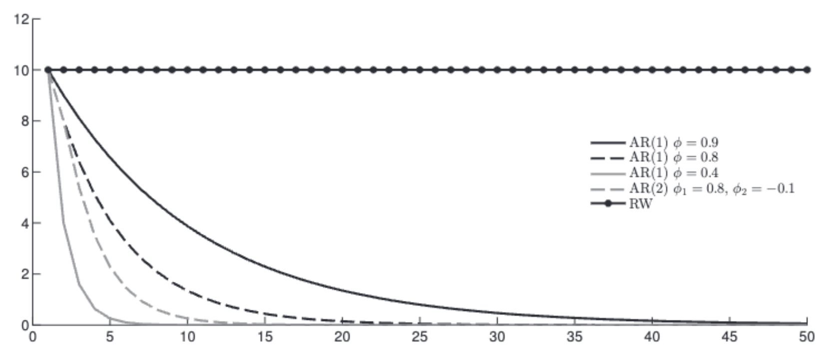
- Best forecast from an RW is:

$$\mathbb{E}[y_{t+1}|y_t] = y_t$$

- The MSE of an RW grows linearly with the forecast horizon:

$$\sigma_y(h) = \text{var}\big(y_{t+h} - \mathbb{E}[y_{t+h}|y_t]\big) = \sigma^2 h$$

- Impulse response function for an RW does not return to zero. Shock persists forever



Note: The figure shows the effect over time of a shock that is normalized to increase all processes with 10 units initially. See the text for details.

- From previous discussions we know that RW is nonstationary. Can also show this by recursive substitution (assuming $y_0 = 0$):

$$\begin{aligned}
y_t &= y_{t-1} + \epsilon_t \\
&= y_{t-2} + \epsilon_{t-1} + \epsilon_t \\
&\vdots \\
y_t &= \sum_{j=0}^{t-1} \epsilon_{t-j} + y_0 \\
y_t &= \sum_{j=0}^{t-1} \epsilon_{t-j}
\end{aligned}$$

- Accordingly, the variance of an RW depends on time:

$$\begin{aligned}
\mathbb{E}[y_t] &= 0 \\
\text{var}(y_t) &= \sigma^2 t
\end{aligned}$$

3.2 Unit root

- The RW model can be made stationary simply by differencing the data:

$$y_t - y_{t-1} = \epsilon_t$$
$$\Delta y_t = \epsilon_t$$

clearly, Δy_t is stationary

Unit root defined: If variable y_t can be made approximately stationary by differencing it once, we say that it is integrated of first order, $I(1)$, or that it has a unit root. Stationary random variables, such as Δy_t are called integrated of order zero, $I(0)$.

3.3 Random Walk with drift

Can also add a constant to the RW:

$$y_t = \alpha + y_{t-1} + \epsilon_t$$

Then we get an RW with drift. The RW with drift can also be represented as a deterministic trend and stochastic term:

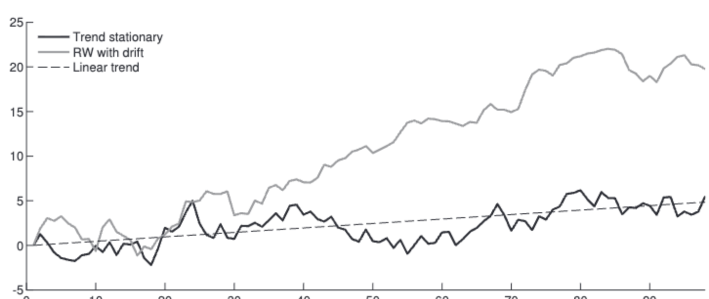
$$\begin{aligned} y_t &= \alpha + y_{t-1} + \epsilon_t \\ &= \alpha + (y_{t-2} + \alpha + \epsilon_{t-1}) + \epsilon_t \\ &= 2\alpha(y_{t-3} + \alpha + \epsilon_{t-3}) + \epsilon_{t-1} + \epsilon_t \\ &\vdots \\ y_t &= \alpha t + \sum_{j=0}^{t-1} \epsilon_{t-j} \end{aligned}$$

with y_0 taken to be zero in the finite past. Compared to the simplest RW model, an RW with drift now also contains a deterministic trend. However, in contrast to the trend stationary model, the deviations from the deterministic trend are not stationary.

3.4 Trend stationarity and difference stationarity

To remember:

- A series that returns to its trend after a shock is called trend stationary
- A series with a unit root is also called difference stationary



Note: We specify $\alpha = 0.05$ in the random walk with drift model $y_t = \alpha + y_{t-1} + \varepsilon_t$. The trend stationary model is simulated as an AR(1) including a deterministic trend: $y_t = \phi y_{t-1} + \alpha t + \varepsilon_t$, where $\phi = 0.8$ and $\alpha = 0.05$. Thus, the slope of the deterministic trend in both models is identical.

4 Implications of Nonstationarity

- Normal "rules" do not apply. i.e., OLS regressions can give spurious results!
- Removing a deterministic trend from a series that has a stochastic model might induce spurious cycles in the data
- If the data has a unit root, we can simply differences the data to make them stationary
- How can we test for unit roots?

5 Tests for Unit Root — The ADF Test

- Many tests do exist. Here: the Augmented Dickey Fuller (ADF) test.
- Consider the following AR(1) process:

$$y_t = \phi y_{t-1} + \epsilon_t \tag{1}$$

  where $\epsilon_t \sim$ i.i.d. $N(0, \sigma^2)$

- The question of interest is whether $\phi = 1$ or $\phi < 1$
  - If $\phi = 1$, equation (1) reduces to an RW
  - If $\phi < 1$, the AR(1) process is stationary

The ADF test:

- (1) can be rewritten as:

$$\Delta y_t = \mu y_{t-1} + \epsilon_t$$

  where $\mu = \phi - 1$

- A test for a unit root is simply a test on the μ parameter, where:

$$H_0 : \mu = 0$$

  which implies that y_t is integrated of order one, $y_t \sim I(1)$, and the alternative:

$$H_1 : \mu < 0$$

  which implies that y_t is stationary, $y_t \sim I(0)$

- Can use t-statistic from the OLS estimate for μ:

$$\hat{t}_{\mathrm{DF}} = \frac{\hat{\phi} - 1}{\mathrm{SE}(\hat{\phi})} = \frac{\hat{\mu}}{\mathrm{SE}(\hat{\mu})}$$

- Note that the ADF test is one-sided and that asymptotic distribution for the t-value is non-Gaussian. i.e., we cannot use the critical values from the standard t-distribution
- It can also be shown that the OLS estimate of μ has a downward bias. Failing to correct for this bias, one is wrongly led to conclude that the series is stationary when it instead is an RW

- Can include more lags to avoid autocorrelation in residuals

$$y_t = \alpha + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$$

which is the same as:

$$\Delta y_t = \alpha + \mu y_{t-1} + \gamma_1 \Delta y_{t-1} + \epsilon_t$$

- Can include deterministic trend to control for drifts:

$$y_t = \alpha + \beta t + \phi y_{t-1} + \epsilon_t$$

which can be rewritten as

$$\Delta y_t = \alpha + \beta t + \mu y_{t-1} + \epsilon_t$$

where $\mu = \phi - 1$, as above. Note that different critical values apply

Variable	**Lags**	**Constant**	**Constant, Trend**
Oslo stock exchange	2	-1.52	-2.57
	4	-1.46	-2.62
	6	-1.32	-2.40
GDP	2	-1.97	-0.45
	4	-1.50	-0.65
	6	-1.66	-1.14
		Critical values	
1%		-3.49	-3.99
5%		-2.88	-3.42
10%		-2.57	-3.14

3-month interest rate	2	-1.25	-2.81
	4	-1.22	-2.99
	6	-1.02	-3.60
Unemployment rate	2	-2.27	-2.19
	4	-2.46	-2.39
	6	-2.66	-2.61
		Critical values	
1%		-3.49	-3.99
5%		-2.88	-3.42
10%		-2.57	-3.14

Note: The Oslo stock exchange, 3-month interest rate and the Trade weighted exchange rate are all monthly variables. CPI, GDP and Unemployment are quarterly variables. The sample is 1981-2013 for all variables, except for Oslo stock exchange, which starts in 1995.

6 Persistence

An alternative method to analyze the degree of nonstationarity has been to analyze for how long macroeconomic variables respond to initial shocks. That is, how persistent are the effects of shocks to the time series?

Persistence implies that a shock lasts for a long time into the future. To measure this we start from the moving average representation of the first difference of a variable y_t.

First, assume y_t is nonstationary, but that the first-differences of the variable are stationary. Then Δy_t has a moving average representation:

$$\Delta y_t = B(L)\epsilon_t = \epsilon_t + \beta_1\epsilon_{t-1} + \beta_2\epsilon_{t-2} + \dots$$

where $\beta_0 = 1$, and as usual, $\epsilon_t \sim$ i.i.d. $N(0, \sigma^2)$. Hence, Δy_t is described entirely by its past and present shocks. $B(L)$ captures the impact of shocks over time. Put differently, the impact (or impulse response) of a shock in period t on the *growth rate* of y in period $t+k$, is B_k. Thus, the impact of the same shock on the *level* of y at time $t+k$ is $1 + B_1 + \cdots + B_k$

Finally, we have learned that the ultimate impact of a shock will be the infinite sum of the moving average coefficients (the cumululative effect of the shock), denoted $B(1)$:

$$\sum_{j=0}^{\infty} \frac{\partial y_{t+j}}{\partial \epsilon_t} = \sum_{j=0}^{\infty} B_j \equiv B(1)$$

The value of $B(1)$ is the measure of persistence, as proposed by Campbell and Mankiw (1987). For an RW, $B(1) = 1$, while for a stationary series (around a deterministic trend), $B(1) = 0$. Hence, we can summarize:

- $B(1) = 1$ for a random walk (shocks persist forever)
- $B(1) > 1$, the ultimate effect is larger than the immediate effect (shocks explode)
- $B(1) < 1$, the ultimate effect is less than the immediate eddect (shocks die out)
- $B(1) = 0$ for a stationary series.

How do we find $B(1)$= Assume Δy_t follows an AR(p) proces:

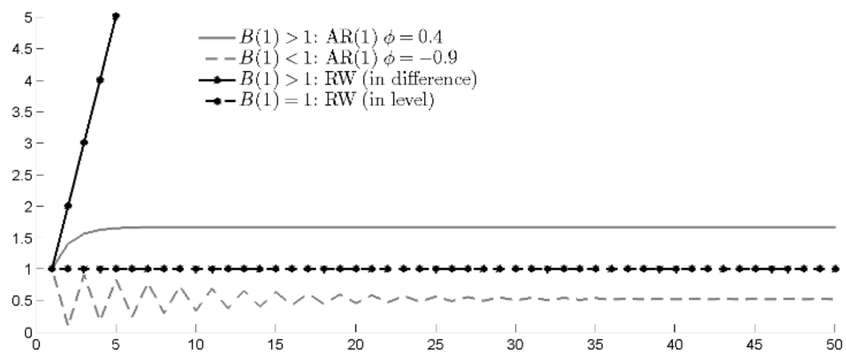
$$\phi(L)\Delta y_t = \epsilon_t$$
$$\Delta y_t = \frac{1}{\phi(L)}\epsilon_t \equiv B(L)\epsilon_t$$

Hence:

$$B(1) = \frac{1}{(1 - \theta_1 - \theta_2 - \cdots - \theta_p)}$$

We have seen before that we can compute $B(1)$ for simple autoregressive models. For example, it was shown that for a stable AR(1), $B(1) = \frac{1}{1-\phi}$. Thus, estimating an AR(1) on Δy_t by OLS, will give us an estimate of ϕ and thereby also $B(1)$.

Figure: Impulse responses for different values of $B(1)$



Note: The figure shows the cumulative (level) effect over time of a shock that is normalized to increase all processes with 1 unit initially. See the text for details.

7 Summary

1. A time series process where either the first- or second-order moments depend on the time index is nonstationary
2. Time series having either a deterministic or stochastic trend are nonstationary
3. If a process returns to its trent (some time) after a shock, we say that it is trend-stationary
4. Time series which are integrated of order one, $I(1)$, can be made stationary by simply differencing the time series. For this reason they are often called difference-stationary. Alternatively we say that they have one unit root
5. The simplest model containing a stochastic trend, and which is difference-stationary, is the random walk
6. Standard regression results as well as time series tools often rely on stationarity. Thus, it is important to test for nonstationarity:
   - Augmented Dickey-Fuller test (or other)
   - Compute degree of persistence